

Trading System Feature List

Here is the complete, final feature list — every single feature we designed, in one place.

Core Architecture

#	Feature	What It Does	How It Works
A1	Multi-Agent CEO Orchestrator	Central decision-maker for all signals	Calls Technical Agent, Institutional Agent, News Agent sequentially. Weighs their reports and fires the final signal. Never fires a trade on its own data.
A2	System A — 5m Technical Engine	Scores chart setup quality (0–100 pts)	7 indicators calculated on closed 5m candles. Generates a score. Sends report to CEO.
A3	System B — Institutional Edge Engine	Reads where smart money is positioned	Processes order flow, volume profile, options chain, and dark pool prints from IBKR tick data. Sends BUY/SELL/Neutral verdict to CEO.
A4	News Intelligence Agent	Tracks global macro calendar and sentiment	Queries Finnhub + FRED + Reuters + Forex Factory + VIX + 10Y Yield. Reports risk level to CEO.
A5	SEC Filing Agent	Auto-discovers institutional market-moving filings	Scans SEC EDGAR every 10 min (day) and 15 min (night). Fires Telegram alerts independently from trade signals.
A6	Position State Machine	Tracks open/closed positions per ticker	Reads/writes positions.json. EXIT signal only fires if an active BUY position exists for that ticker.

Data Layer

#	Feature	What It Does	How It Works
D1	IBKR WebSocket Live Feed	Real-time candle and tick data	Connects to IBKR Paper account via ib_insync. Subscribes to 5m bar stream + tick-by-tick stream per ticker. Event-driven — signals fire on candle close, not a timer.
D2	IBKR Historical Data (Backtest)	Clean 90-day bar history for testing	Pulls and caches 90 days of 5m bars from IBKR to local files. Re-used for backtest without repeated API calls. Avoids yfinance rate-limit and data quality issues.
D3	Rate Limit Manager	Prevents IBKR API throttling	Staggers historical data requests to max 6 per 10 seconds. Options chain queries only fire when System A score exceeds 50.
D4	Market Hours Gate	System only runs during US regular session	Enforces 09:25–16:05 ET, Monday–Friday. System sleeps outside hours. SEC Agent and EOD report are exceptions.

System A — 7 Indicators

#	Indicator	Points	Entry Trigger	Why This Indicator
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S1	VWAP Bounce	20 pts	Price retraces to [VWAP, VWAP + 0.15% ATR] and prints a bullish candle	Institutions execute block orders at VWAP. Bouncing off it = real demand, not a fake-out
S2	EMA 9/21 Ribbon	20 pts	EMA 9 crosses above EMA 21 within the last 2 candles	Faster than SMA, catches momentum shifts early. Ribbon width measures acceleration velocity
S3	MACD Histogram Slope	15 pts	Histogram rises for 3 consecutive bars	Measures rate-of-change, not just direction. Catches momentum build-up before the actual price breakout
S4	RSI Hidden Divergence	15 pts	Price makes higher low while RSI makes lower low	Exposes seller exhaustion. Much more reliable than simple RSI < 30 (which fires constantly in downtrends)
S5	Volume Delta Confirmation	15 pts	Current candle volume > 1.5x 20-bar average	Filters fake-outs. If no volume, institutions aren't participating — ignore the move
S6	Bollinger Band Squeeze	10 pts	BB compresses inside Keltner Channels, then expands	Volatility cycles from compressed to explosive. Catches breakout entries before the move happens
S7	Order Block Proximity	5 pts	Price within 0.3% of last down-close candle before a prior swing up	Identifies exact price levels where institutional limit orders are waiting

Higher Timeframe Context Modifier (applied on top of score):

1D	1H	Modifier
Bullish	Bullish	+20 pts (ride the trend)
Bullish	Bearish	+10 pts (dip in uptrend)
Bearish	Bullish	0 pts (neutral)
Bearish	Bearish	-10 pts (counter-trend — fires but flags warning)

System B — Institutional Components

#	Component	What It Reads	Trading Meaning
I1	Volume Delta	Aggressive buys vs sells per 5m candle	Confirms who controls the candle — buyers or sellers
I2	Cumulative Delta	Running session total of buy/sell pressure	Shows who controls the entire day — sustained institutional buying or selling
I3	DOM Imbalance	Top 10 bid depth vs top 10 ask depth	Bid stack ≥ 3x ask = institutional buy wall present below price
I4	Volume Profile (POC/VAH/VAL)	Volume-at-price for the current session	Price returning to POC = high-probability bounce. Low volume nodes = price moves through fast
I5	Options Chain Activity	Put/Call ratio + unusual strike volume	P/C ratio spike = hedging. Call volume > 2.5x open interest at a strike = institutional positioning
I6	Dark Pool Approximation	Block trades > 10k shares at bid/ask midpoint	Invisible institutional accumulation visible on tape as block mid-price crosses

Signal Types & Entry Mechanics

#	Signal Type	Triggers When	Position Size	Stop-Loss	Targets
Trend					
E1	Continuation BUY	System A ≥ 80 AND System B = BUY	Full	1.5x ATR (locked, trails up)	Target 1 + Target 2
E2	Momentum Scalp BUY	System A 65–79 OR System A ≥ 80 + System B Neutral	Half	1.0x ATR (locked, trails up)	Target 1 only (auto-exit)
E3	Conflict Info Alert	System A ≥ 80 BUT System B = SELL	No trade	N/A	No trade — logged for review
E4	Emergency EXIT	Price Guardian detects stop breach mid-candle	Close position	N/A	Immediate Telegram alert

Entry Zone Calculation: [Current Close, Current Close + 0.25% $\times$ ATR] — prevents chasing breakouts.

Exit Engine

#	Feature	Trend Continuation	Momentum Scalp
X1	MACD Decline	3 consecutive bars	2 consecutive bars
X2	Volume Climax	Volume $> 2.5\times$ avg AND close in bottom 30% of candle range	Volume drops to $< 50\%$ of avg (exhaustion)
X3	RSI Momentum Break	RSI falls below 50 after crossing above 60	N/A
X4	EMA Structure Break	Price closes below EMA 9 for 2 consecutive candles	N/A
X5	Trailing Stop Breach	ATR multiplier 1.5x — ratchets up only	ATR multiplier 1.0x — ratchets up only
Rule		2 of 5 conditions must trigger	1 of 4 conditions must trigger

Trailing Stop Formula: Stop_t = Max(Stop_{prev}, Close_t – (ATR $\times$ Multiplier)) — never moves down.

Risk Management

#	Feature	Trigger	Action
R1	Price Guardian	Price $<$ stop-loss OR drop $> 1.5\%$ in < 3 min	Immediate Telegram emergency exit. No candle close needed
R2	VIX Spike Gate	VIX > 25	All scalp signals halted. Score penalty –20 on all signals
R3	VIX Hard Halt	VIX > 35	All signals fully frozen until VIX drops
R4	Circuit Breaker L1	2 consecutive daily losses	Warning alert sent to Telegram
R5	Circuit Breaker L2	3 consecutive daily losses	All scores –20. Only Dual Confirm signals fire
R6	Circuit Breaker L3	4 losses OR daily P&L $< -2\%$	Complete system halt for remainder of session
R7	Signal Staleness Timer	5 min / 10 min / 15 min after signal fire	Follow-up alerts showing current price. Auto-expires position state at 15 min
R8	Pre-Event Compression Detector	Major event within 72 hours AND ATR declining 3 days	Score penalty –15. Signal flagged "Pre-Event Compression"

Macro Intelligence Layer

#	Source	What It Monitors	Action
M1	Finnhub Calendar	Global scheduled events with importance rating	Triggers 3-phase macro block on HIGH-importance events
M2	FRED API	Official US: CPI, NFP, PCE, GDP, Retail Sales	Pre-event block 30 min before, 15 min post-event fake-out window
M3	Forex Factory RSS	ECB, BOJ, BOE, China PMI, global central banks	Sector-level score penalty for affected assets
M4	Reuters RSS	Geopolitical news	Sentiment scoring, reduces scores if global risk-off detected
M5	VIX + 10Y Yield	Live market fear gauge and rate pressure	Hard gates (see Risk Management)
M6	Gold (GLD) Monitor	Risk-off indicator	Context label on signals — does not gate signals
M7	DXY Dollar Index	Strong dollar = pressure on mega-cap earnings	Context label on signals — does not gate signals
M8	3-Phase Macro Block	Pre-event / Fake-out / Reclaim phases	Blocks entries pre/during event. Enables special "Post-Event Opportunity" signal after VWAP reclaim

SEC Filing Agent

#	Filing Type	What It Means	Alert Sent
F1	Form 4	Corporate insider buying > \$1M	Immediately to Telegram with name, amount, ticker
F2	SC 13D	Activist investor accumulating > 5% stake	Immediately — often precedes major price move
F3	8-K (M&A events)	Merger, acquisition, material event	Immediately — flag for potential gap trade
F4	Auto-discovery	Detects hot tickers NOT in your watchlist	Alert with suggestion to add to watchlist
Schedule	Day: every 10 min	After-hours: every 15 min (until 11 PM ET)	Weekends: 10 AM Saturday + Sunday sweep

Telegram Alerts

#	Alert Type	When Sent	Key Details
T1	BUY Signal	New entry signal fires	Ticker, signal type, entry zone, Target 1, Target 2, stop-loss, R:R ratio, score breakdown, time quality, macro status
T2	EXIT Signal	Position close conditions met	Ticker, entry price, exit price, hold time, P&L %, exit reason (which conditions triggered)
T3	Conflict Report	A ≥ 80 but B = SELL	Why trade was skipped, both system readings, logged for EOD validation
T4	Emergency EXIT System	Price Guardian mid-candle breach	Immediate — no wait for candle close

T5	Health Alert	Connection lost / data issue	Last known signal state, retry status, manual action instructions
T6	Daily Heartbeat	Every hour during session (if quiet)	System status, active connections, next macro event
T7	Signal Staleness Good	5/10/15 min after signal fire	"Signal is 8 min old. Entry zone may have passed. Current price: \$X"
T8	Morning Alert	09:25 AM ET daily	Today's macro calendar, VIX level, pre-market gap summary
T9	SEC Filing Alert	After any material SEC filing detected	Filing type, company, insider name, dollar amount, link

Reporting & Analysis

#	Report	When	What It Contains
P1	End-of-Day Summary	4:05 PM ET daily	All signals, wins/losses, P&L, System A vs B accuracy, circuit breaker usage
P2	Tomorrow's Plan	4:05 PM ET daily	Tomorrow's macro calendar, compression risk, watchlist gap zones, institutional positioning
P3	Weekly Performance Report	Friday 4:30 PM ET	Weekly win rate, best/worst setups, indicator performance, auto-tuning suggestions
P4	Conflict Log Report	Included in EOD	How many conflict skips — how many would have won/lost. CEO validation score.

Backtest Laboratory (UI)

#	Feature	Detail
B1	90-Day Backtest	Uses cached IBKR historical data. 1 ticker at a time. Runs as background task — doesn't block UI
B2	System A vs B vs Consensus	Shows win rate, avg win, avg loss, EV per trade, max drawdown, Sharpe ratio for each system independently
B3	Signal History Table	Filterable table of all historical signals: date, system, type, P&L, win/loss
B4	Signal Replay	Click any historical signal → see exact indicator state at entry. Teaches pattern recognition

User Interface

#	Feature	Detail
U1	Watchlist Selector	Searchable dropdown, up to 10 tickers (max 15), pre-loaded with all US stock symbols
U2	Score Threshold Slider	Adjust 65 (moderate) and 80 (high conviction) thresholds live
U3	Time Quality Labels	Every signal tagged: Prime Time / Low Volume △ / Open Chaos
U4	Metrics-Only UI	Clean cards per ticker showing: price, score, signal type, VWAP, EMA, stop-loss, position state

U5	Position State Display	Shows OPEN / FLAT per ticker with entry price and current P&L
U6	Macro Status Banner	Top-of-page bar showing: next event, VIX level, 10Y yield, system health

Security & Performance

#	Feature	Detail
SC1	Secrets Management	Telegram token, user IDs in st.secrets (.streamlit/secrets.toml) — gitignored, never in code
SC2	Local JSON Storage	No cloud DB. positions.json, signal_history.jsonl, conflict_log.jsonl — survives restarts
SC3	API Rate Guards	Each external API call has a hardcoded minimum interval enforced by scheduler
SC4	Memory Management	Session state cleaned at 4:05 PM daily. DataFrames use rolling windows, not growing history
SC5	Reconnection Logic	IBKR disconnect → 3 retries with exponential backoff → Telegram alert if all fail
SC6	Weekend/Night Sleep	System fully idles 11 PM – 6 AM. SEC Agent runs Saturday/Sunday 10 AM sweep only

Total Features: 65 across 10 categories. Everything discussed, nothing missing.